Project Proposal

New Lab Access System for Sandboxx

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The System in Place and Associated Problems

Currently, the access is regulated using a lock which uses fingerprints to verify users. The authority to create new users lies with the IIC office. This space is used by 6 hardware technical clubs with total members exceeding 200. Ideally, Sandboxx should be accessible to all but since this is an unsupervised space with a lot of freedom, it is also prone to misuse and unintended theft and damage. Naturally, we want some method of accountability in place.

Over the last few years, certain operating models have been tested. When access was given to any project/team which was working over there, the above-mentioned ill-effects were prevalent. In the year when every club had one access holder (total 6 in the entire student body), the space started being used properly. But this model inherently has some friction in operation as anyone who wants to enter Sandboxx requires one of these access holders present; be it 11pm in the night or 10am in the morning. Since access holders have no benefit in walking over from their hostels and opening Sandboxx for things they are not involved in, naturally the space is not utilized at its potential.

Proposed Solution

The lock will regulate access using 6-digit pins which are generated on the server (DADU). The lock is connected to the local lan via WIFI from esp32 module. There are 3 roles in this system: Admin (Professors, IIC, etc), SuperUsers, and users. All are registered and authenticated via their bits-mail. The users, upon registration, are asked which Superuser they would want to be associated with. They can be associated with multiple SuperUsers. Whenever the user wants to access Sandboxx, they are asked for the reason in brief and then click on their dashboard to reveal the generated pin. This pin is valid for 12 hours (Subject to change upon requirement). The pin is only revealed upon the confirmation of the SuperUser. The SuperUser's phone number is available on the user's dashboard.

The SuperUsers are composed of 2nd years or 3rd years (in a case of exception). They are registered and removed to the platform by the Admin. The SuperUsers and the Admin can view the logs. Users and SuperUsers are deleted annually.